

OpenArt.ai Enhanced Environment Integration

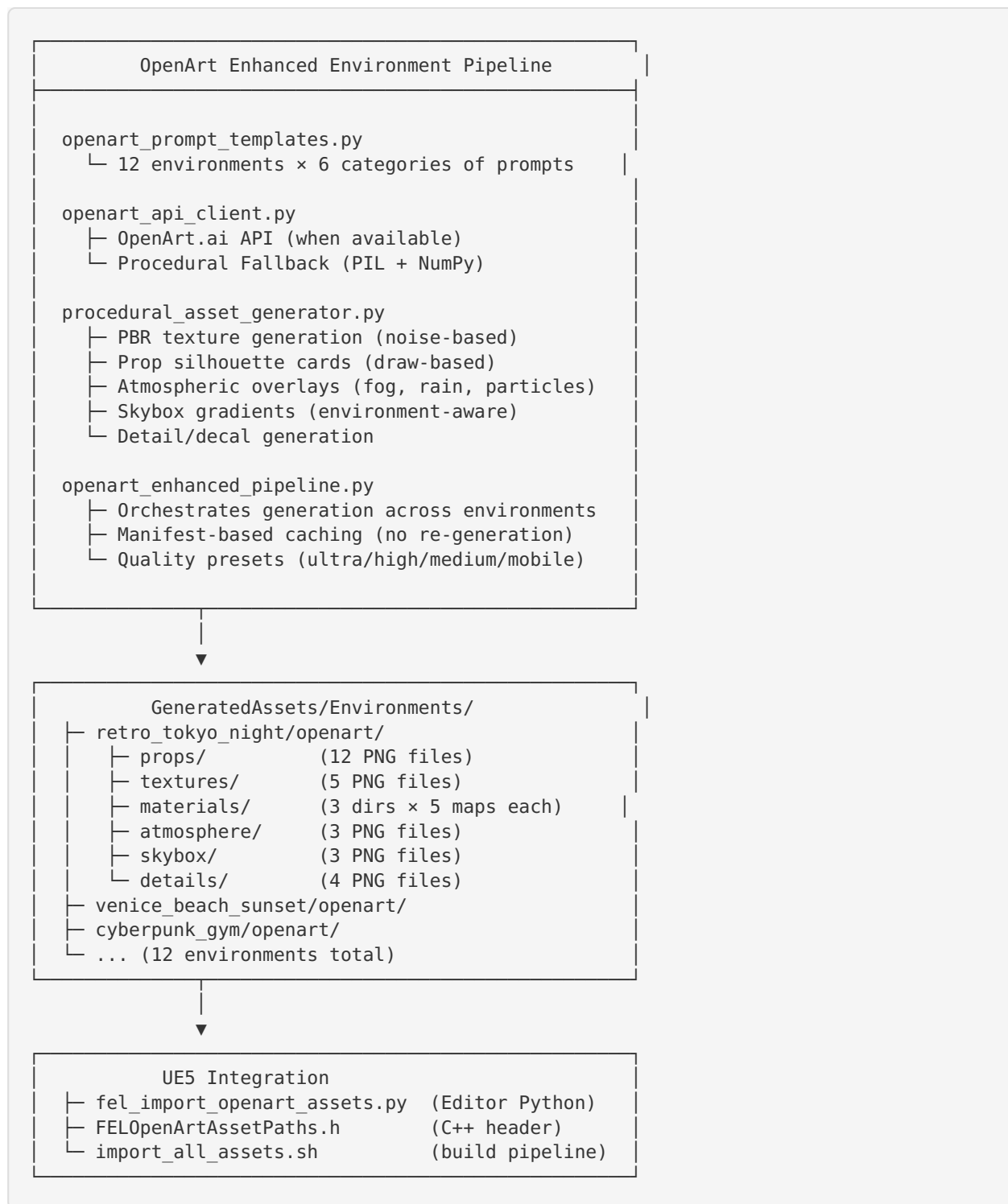
Overview

The OpenArt integration adds **504 additional world assets** across all **12 preset environments** in Final Evolution Lab. These assets enhance visual quality by adding detailed props, PBR material sets, atmospheric effects, skybox variations, and detail decals to every scene.

Asset Summary

Category	Per Environment	Total (12 envs)	Description
Props	12	144	Scene objects (trees, signs, furniture, equipment)
Textures	5	60	Tileable surface textures (walls, floors, ground)
Materials	3 × 5 maps	180 files	PBR material sets (albedo, normal, roughness, metallic, AO)
Atmosphere	3	36	Fog, particles, light rays, weather effects
Skybox	3	36	Equirectangular panoramic sky variations
Details	4	48	Graffiti, stains, cracks, decorative elements
Total	30 entries	504 files	

Architecture



Quick Start

Generate Assets

```
cd /path/to/rork-final-evolution-lab

# Generate all environments at high quality
python -m scripts.environment_gen.openart_enhanced_pipeline --quality high

# Generate a single environment
python -m scripts.environment_gen.openart_enhanced_pipeline --env retro_tokyo_night

# Generate only textures for all environments
python -m scripts.environment_gen.openart_enhanced_pipeline --category textures

# Preview without generating
python -m scripts.environment_gen.openart_enhanced_pipeline --dry-run

# List all environments and asset counts
python -m scripts.environment_gen.openart_enhanced_pipeline --list
```

Import into UE5

```
# Via the full build pipeline
./scripts/ue5_setup/fel_complete_pipeline.sh

# Or import OpenArt assets only (in UE5 Editor)
# Edit → Editor Preferences → Python → Execute:
#   EditorPython/fel_import_openart_assets.py

# Or via commandlet
$UE_CMD $FEL_PROJECT -run=pythonscript \
  -script=EditorPython/fel_import_openart_assets.py \
  -nosplash -unattended -nopause
```

Environment Details

1. Retro Tokyo Night

Neon-soaked Tokyo street court at night with rain reflections, graffiti, and vending machines.

- **Props:** Neon signs, vending machines, phone booths, ramen carts, shrine gates
- **Textures:** Wet asphalt, brick alleys, metal shutters
- **Atmosphere:** Neon fog, rain particles, volumetric light rays

2. Venice Beach Sunset

California beach court at golden hour with palm trees and skatepark vibes.

- **Props:** Palm trees, lifeguard tower, skateboard ramp, surfboard rack, taco truck
- **Textures:** Beach sand, boardwalk wood, stucco walls
- **Atmosphere:** Golden haze, dust in sunlight, sunset god rays

3. Cyberpunk Gym

Futuristic training facility with holograms, neon lighting, and tech equipment.

- **Props:** Hologram displays, training pods, LED floor strips, drone cameras

- **Textures:** Hexagonal tiles, carbon fiber, circuit board walls
- **Atmosphere:** Cyan/magenta haze, holographic particles, neon bloom

4. Classic NBA Arena

1990s-style basketball arena with hardwood court and packed crowd.

- **Props:** Jumbotron, courtside seats, championship banners, shot clocks
- **Textures:** Maple hardwood, painted lanes, arena seat fabric
- **Atmosphere:** Spotlight haze, crowd dust, dramatic light beams

5. Underground Bunker

Military-style underground training facility with raw concrete and industrial lighting.

- **Props:** Heavy bags, ammo crates, tire stacks, military lockers
- **Textures:** Raw concrete, rusted metal, cinder block, rubber mats
- **Atmosphere:** Chalk dust, condensation fog, harsh spotlights

6. Rooftop Cityscape

Urban rooftop court above a megacity skyline at dusk.

- **Props:** Chain-link fence, AC units, water towers, fire escapes
- **Textures:** Tar rooftop, brick parapet, metal grating
- **Atmosphere:** City haze, wind dust, sunset rays between buildings

7. Winter Outdoor

Outdoor court in winter with snow, frost, and warm street lamp glow.

- **Props:** Bare trees, frozen hoops, fire barrels, snow piles
- **Textures:** Snow ground, icy concrete, frozen asphalt
- **Atmosphere:** Snowfall particles, breath fog, warm lamp glow

8. Beach Tropical

Paradise island court on white sand with turquoise water.

- **Props:** Coconut palms, beach umbrellas, tiki torches, hammocks
- **Textures:** White sand, palm bark, bamboo mat, coral stone
- **Atmosphere:** Ocean mist, sun sparkle, tropical light rays

9. Warehouse Industrial

Converted warehouse gym with exposed brick and industrial fixtures.

- **Props:** Steel beams, cage lights, pallet stacks, boxing ring, forklifts
- **Textures:** Exposed brick, concrete, corrugated metal, wood planks
- **Atmosphere:** Dust haze, smoke wisps, skylight shafts

10. Neon Arcade

Synthwave 80s arcade with glowing grid floors and vaporwave aesthetics.

- **Props:** Arcade cabinets, pinball machines, disco balls, neon palms
- **Textures:** Neon grid floor, pixel tiles, chrome panels
- **Atmosphere:** Neon haze, sparkle particles, laser beams

11. Zen Dojo

Traditional Japanese training hall with tatami, wood, and serene minimalism.

- **Props:** Bonsai trees, paper screens, weapon racks, incense holders

- **Textures:** Tatami mats, dark wood beams, washi paper, zen garden sand
- **Atmosphere:** Incense smoke, dust in sunlight, warm lantern glow

12. Stadium Night Game

Professional soccer stadium under floodlights with packed crowd.

- **Props:** Floodlight towers, goal posts, corner flags, camera cranes
- **Textures:** Green grass pitch, running track, concrete terraces
- **Atmosphere:** Floodlight haze, crowd flare smoke, firework sparks

OpenArt API Integration

Current Status

OpenArt.ai does **not currently expose a public REST API**. The system is designed to seamlessly switch to the API when it becomes available.

When API is Available

1. Set `OPENART_API_KEY` in `.env`
2. The client will automatically detect and use the API
3. All prompt templates are optimized for AI image generation
4. Polling-based task completion handles async generation

Procedural Fallback

When the API is unavailable, assets are generated procedurally:

- **Textures:** Fractal Brownian Motion noise with environment-specific color palettes
- **Normal maps:** Gradient-based height → normal conversion
- **Roughness/Metallic:** Material-aware value ranges
- **Props:** Silhouette cards with themed shapes (trees, signs, furniture)
- **Atmosphere:** Layered noise fog, particle overlays, volumetric ray approximations
- **Skybox:** Multi-stop gradient with cloud noise and star patterns

All procedural assets are **deterministic** — the same prompt always produces the same result, enabling caching and reproducibility.

Quality Presets

Preset	Texture	Prop	Material	Skybox	Target
ultra	2048	1024	2048	4096	High-end PC
high	1024	512	1024	2048	PC / Console
medium	512	512	512	1024	Mid-range PC
mobile	512	256	512	1024	iOS / Android

File Structure

```

scripts/environment_gen/
├── openart_api_client.py           # API client with procedural fallback
├── openart_prompt_templates.py    # 360 prompt definitions for 12 environments
├── openart_enhanced_pipeline.py  # Main generation orchestrator (CLI)
└── procedural_asset_generator.py  # PIL/NumPy procedural generation engine

GeneratedAssets/Environments/
├── openart_asset_manifest.json    # Master manifest tracking all 504 assets
├── <env_key>/openart/
│   ├── props/                   # 12 prop PNG files
│   ├── textures/                # 5 texture PNG files
│   ├── materials/               # 3 PBR material directories
│   └── <material_name>/        # 5 PBR maps each (albedo, normal, roughness,
│       │                       metallic, ao)
│       ├── atmosphere/          # 3 atmospheric overlay PNG files
│       ├── skybox/              # 3 equirectangular skybox PNG files
│       └── details/             # 4 detail/decal PNG files

UnrealStarter/BasketballGame/
├── EditorPython/
│   └── fel_import_openart_assets.py # UE5 batch import script
├── Source/FinalEvolutionLab/
│   └── FELOpenArtAssetPaths.h      # C++ asset path constants (360 entries)

```

Troubleshooting

“No assets generated”

- Ensure Python 3.8+ with `Pillow` and `numpy` installed
- Check that `GeneratedAssets/Environments/` directory exists
- Run with `--list` flag to verify prompt template loading

“Material instances not created in UE5”

- Ensure base material `M_OpenArt_PBR` exists at `/Game/FEL/Materials/`
- Run import script inside UE5 Editor (not commandlet)

“Manifest says cached but files missing”

- Delete `openart_asset_manifest.json` to force regeneration
- Or delete specific environment directory and re-run

Performance

- Generation time: ~8 minutes for all 504 assets (medium quality)
- Disk usage: ~60-80 MB total (all environments)
- Memory: Peak ~200 MB during generation